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Industrial Internship Report on

UrbanFlowAI

Smart City Traffic Forecasting using Machine Learning

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(The Northcap University)

Executive Summary
This report provides details of the Industrial Internship provided by upskill Campus and The IoT Academy in collaboration with Industrial Partner UniConverge Technologies Pvt Ltd (UCT). This internship was focused on a project/problem statement provided by UCT. We had to finish the project including the report in 6 weeks' time. My project was "UrbanFlow AI – Smart City Traffic Forecasting using Machine Learning," an intelligent web-based application developed to predict traffic congestion levels using historical traffic and environmental data. The project leverages Machine Learning algorithms to analyze factors such as temperature, weather conditions, holidays, day of the week, hour of the day, and month to forecast traffic volume accurately. A user-friendly dashboard was built using Streamlit, allowing users to select different cities and visualize traffic predictions along with interactive charts and analytics. The application also features dynamic city background images, real-time prediction capability, and an enterprise-style interface, making it both practical and visually engaging. This internship gave me a very good opportunity to get exposure to Industrial problems and design/implement solution for that. It was an overall great experience to have this internship.

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1.Preface

The six-week Industrial Internship organized by **upskill Campus (USC)** in collaboration with **UniConverge Technologies (UCT)** and **The IoT Academy** provided me with an excellent opportunity to gain practical exposure to modern technologies and industry-oriented project development. This internship helped me bridge the gap between theoretical concepts learned in academics and their real-world implementation through hands-on experience in Machine Learning, Python programming, data analysis, and web application development.

As part of this internship, I developed **"UrbanFlow AI – Smart City Traffic Forecasting using Machine Learning,"** an intelligent web-based application designed to predict traffic congestion using Machine Learning techniques. The system analyzes various parameters such as temperature, weather conditions, holidays, day of the week, hour, and month to estimate traffic levels. A modern dashboard was developed using Streamlit, featuring interactive visualizations, city-specific backgrounds, analytical charts, and real-time traffic predictions. The project demonstrates how Artificial Intelligence can contribute to efficient traffic management and support smart city initiatives.

The internship program was systematically planned with structured learning modules, practical assignments, project implementation, documentation, and GitHub-based project submission. Throughout these six weeks, I enhanced my knowledge of Python, Machine Learning, data preprocessing, visualization techniques, Streamlit application development, Git and GitHub, and technical documentation. In addition to technical skills, I also improved my logical thinking, debugging ability, project management, and self-learning capabilities.

I would like to express my sincere gratitude to **upskill Campus**, **UniConverge Technologies Pvt. Ltd.**, and **The IoT Academy** for providing this valuable learning opportunity. I am thankful to all the mentors, trainers, coordinators, faculty members, and my college for their continuous guidance and support throughout the internship.

Finally, I would like to encourage my juniors and peers to actively participate in internship programs and continuously explore new technologies.

2.Introduction

2.1 About UniConverge Technologies Pvt Ltd

A company established in 2013 and working in Digital Transformation domain and providing Industrial solutions with prime focus on sustainability and RoI.

For developing its products and solutions it is leveraging various **Cutting Edge Technologies e.g. Internet of Things (IoT), Cyber Security, Cloud computing (AWS, Azure), Machine Learning, Communication Technologies (4G/5G/LoRaWAN), Java Full Stack, Python, Front end** etc.

i. UCT IoT Platform (uct Insight)

UCT Insight is an IOT platform designed for quick deployment of IOT applications on the same time providing valuable "insight" for your process/business. It has been built in Java for backend and ReactJS for Front end. It has support for MySQL and various NoSql Databases.

- It enables device connectivity via industry standard IoT protocols - MQTT, CoAP, HTTP, Modbus TCP, OPC UA
- It supports both cloud and on-premises deployments.

It has features to
• Build Your own dashboard
• Analytics and Reporting
• Alert and Notification
• Integration with third party application(Power BI, SAP, ERP)
• Rule Engine

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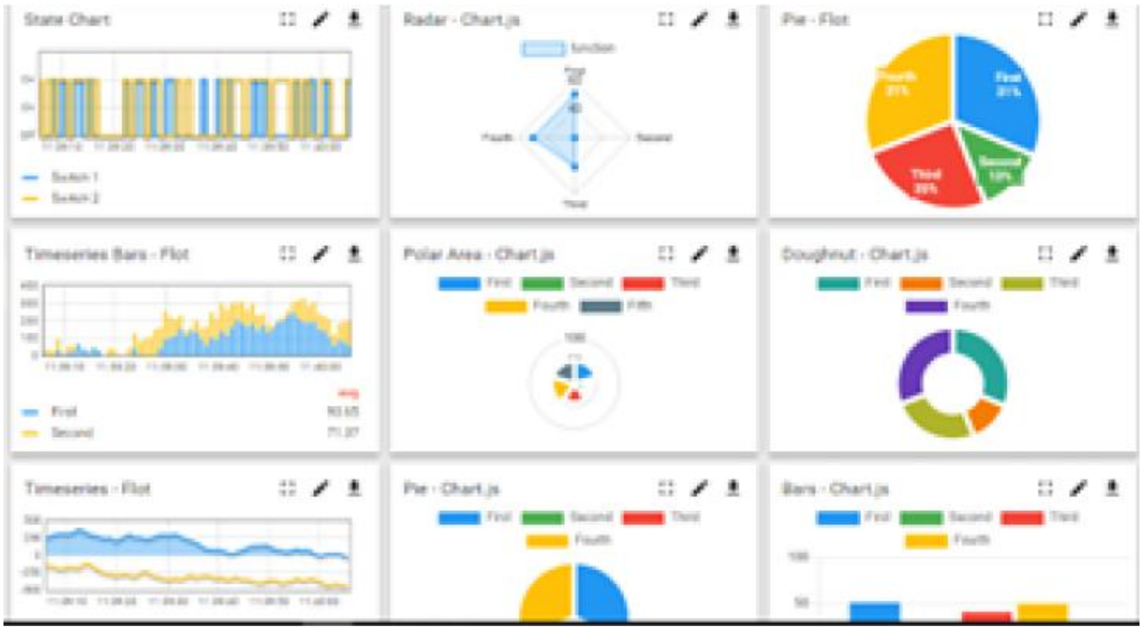

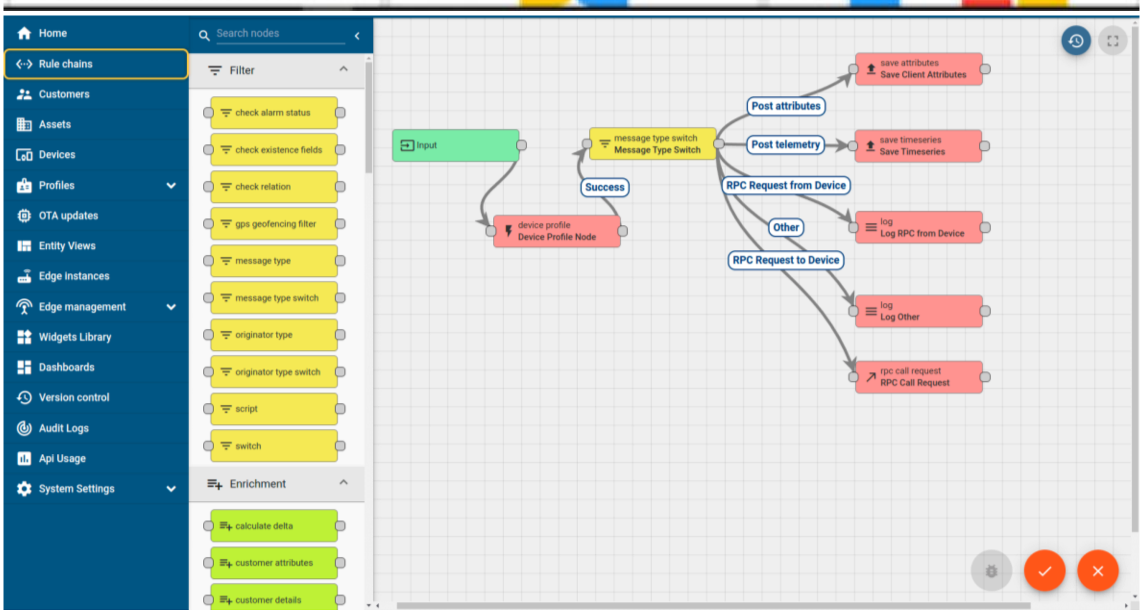

ii. Smart Factory Platform (FACTORY WATCH)

Factory watch is a platform for smart factory needs.

It provides Users/ Factory

- with a scalable solution for their Production and asset monitoring
- OEE and predictive maintenance solution scaling up to digital twin for your assets.
- to unleased the true potential of the data that their machines are generating and helps to identify the KPIs and also improve them.
- A modular architecture that allows users to choose the service that they what to start and then can scale to more complex solutions as per their demands.

Its unique SaaS model helps users to save time, cost and money.

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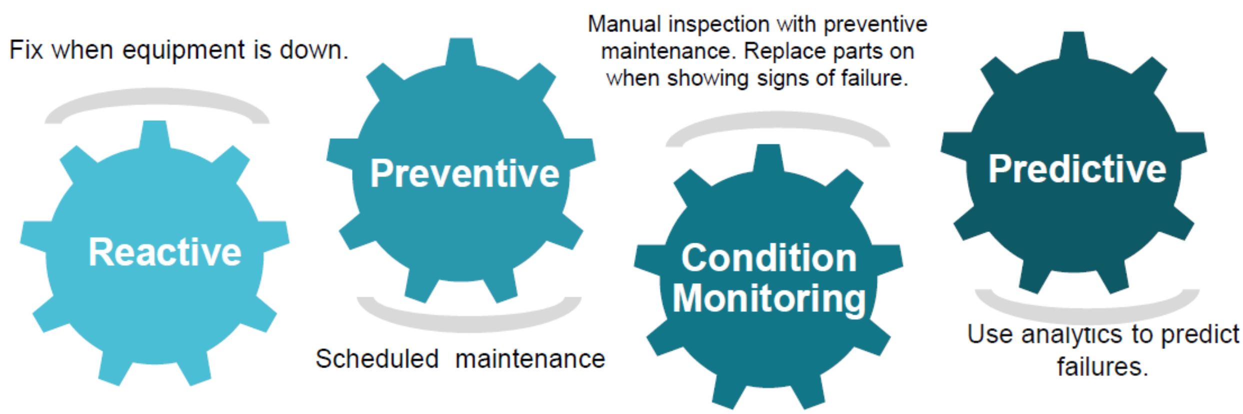
Machine	Operator	Work Order ID	Job ID	Job Performance	Job Progress		Output		Rejection	Time (mins)				Job Status	End Customer
					Start Time	End Time	Planned	Actual		Setup	Pred	Downtime	Idle		
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i

iii. LoRaWAN based Solution

UCT is one of the early adopters of LoRAWAN teschnology and providing solution in Agritech, Smart cities, Industrial Monitoring, Smart Street Light, Smart Water/ Gas/ Electricity metering solutions etc.

iv. Predictive Maintenance

UCT is providing Industrial Machine health monitoring and Predictive maintenance solution leveraging Embedded system, Industrial IoT and Machine Learning Technologies by finding Remaining useful life time of various Machines used in production process.



2.2 About upskill Campus (USC)

upskill Campus along with The IoT Academy and in association with Uniconverge technologies has facilitated the smooth execution of the complete internship process.

USC is a career development platform that delivers **personalized executive coaching** in a more affordable, scalable and measurable way.

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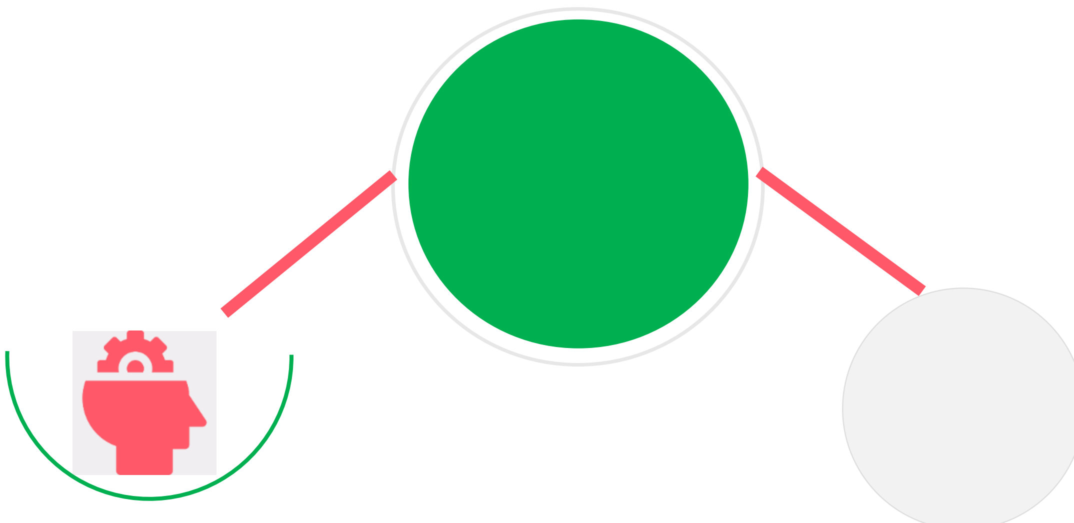
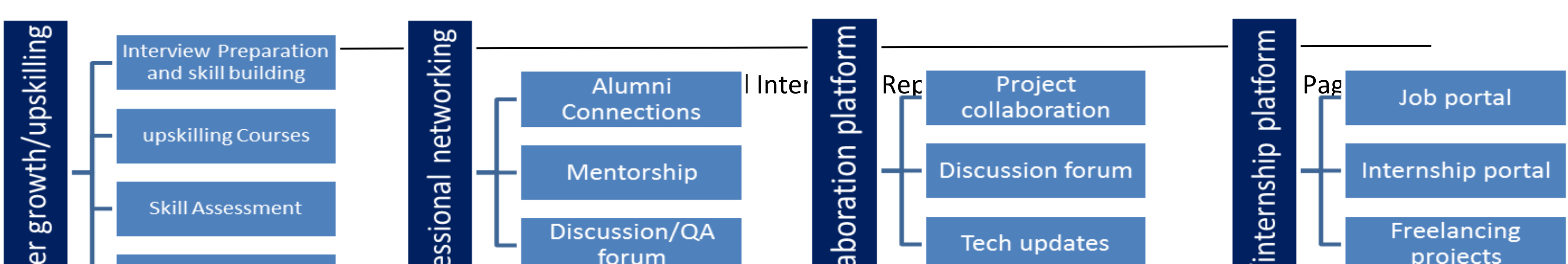


Seeing need of upskilling in self paced manner along-with additional support services e.g. Internship, projects, interaction with Industry experts, Career growth Services

upSkill Campus aiming to upskill 1 million learners in next 5 year

https://www.upskillcampus.com/



2.3 The IoT Academy

The IoT academy is EdTech Division of UCT that is running long executive certification programs in collaboration with EICT Academy, IITK, IITR and IITG in multiple domains.

2.4 Objectives of this Internship program

The objective for this internship program was to

☛ get practical experience of working in the industry.

☛ to solve real world problems.

☛ to have improved job prospects.

☛ to have Improved understanding of our field and its applications.

☛ to have Personal growth like better communication and problem solving.

2.5 Reference

[1] Python Software Foundation, **Python Documentation**, https://docs.python.org/

[2] Streamlit Inc., **Streamlit Documentation**, https://docs.streamlit.io/

[3] Scikit-learn Developers, **Scikit-learn Documentation**, https://scikit-learn.org/

2.6 Glossary

Terms	Description
Random Forest	A Machine Learning algorithm that combines multiple decision trees to improve prediction accuracy.
Machine Learning	A branch of AI where computers learn patterns from data to make predictions.
API	A set of rules that allows different software applications to communicate with each other.
Feature Engineering	The process of selecting and transforming input variables to improve Machine Learning model performance.
Streamlit	A Python framework used to build interactive web applications.
Model Training	The process of teaching a Machine Learning model using historical data to recognize patterns and make predictions.

3.Problem Statement

SmartCityTrafficForecasting

With the rapid growth of urban populations and the increasing number of vehicles on roads, traffic congestion has become one of the major challenges faced by modern cities. Traffic jams lead to increased travel time, higher fuel consumption, air pollution, road accidents, and economic losses. Traditional traffic monitoring systems mainly provide current traffic information but often lack the ability to predict future traffic conditions accurately, making proactive traffic management difficult.

The objective of the **SmartCityTrafficForecasting** project is to develop an intelligent Machine Learning-based system capable of predicting traffic volume using historical traffic and environmental data. The system analyzes important factors such as temperature, weather conditions, holidays, day of the week, hour of the day, and month to estimate future traffic levels. By identifying traffic patterns in advance, the system helps authorities and commuters make informed decisions regarding traffic management, route planning, and resource allocation.

To address this problem, a Machine Learning model based on the **Random Forest Regressor** algorithm was developed and integrated with an interactive **Streamlit** dashboard. The application provides real-time traffic predictions, visual analytics, and dynamic city-specific interfaces, enabling users to understand traffic trends easily. The proposed solution demonstrates how Artificial Intelligence and Machine Learning can be applied to improve urban transportation systems and contribute to the development of smarter, safer, and more efficient cities.

4.Existing and Proposed solution

Existing Solution

Traffic management systems currently used in many cities primarily rely on GPS navigation, traffic cameras, and sensor-based monitoring to provide real-time traffic information. Popular navigation applications such as Google Maps and Waze use live user data and GPS signals to estimate current traffic conditions and suggest alternative routes. Intelligent Transportation Systems (ITS) also use cameras, IoT sensors, and traffic control centers to monitor vehicle movement and manage traffic flow.

Although these systems are effective in displaying current traffic conditions, they have several limitations:

- They mainly focus on **real-time monitoring** rather than predicting future traffic conditions.
- Most systems require continuous internet connectivity and access to live GPS or sensor data.
- They often involve expensive infrastructure such as cameras, sensors, and IoT devices.
- They provide limited insights into how environmental factors such as weather, holidays, or seasonal variations influence traffic patterns.
- Many existing solutions do not offer customizable dashboards for detailed traffic analytics and visualization.

Due to these limitations, there is a need for an intelligent system capable of forecasting traffic conditions using historical data and Machine Learning techniques.

Proposed Solution

- To overcome the limitations of existing systems, the **SmartCityTrafficForecasting** project proposes an AI-powered traffic prediction system using Machine Learning. The application is designed to predict traffic volume based on historical traffic data and multiple influencing factors such as temperature, weather conditions, holidays, day of the week, hour of the day, and month.
- The project uses the **Random Forest Regressor** algorithm to train the prediction model and provides an interactive web dashboard developed using **Streamlit**. Users can enter traffic-related parameters, select a city, and instantly receive predicted traffic levels along with visual analytics. The dashboard also includes interactive graphs, city-specific background images, and an intuitive user interface, making it suitable for traffic analysis and demonstration purposes.

Value Addition

The proposed system offers several improvements over conventional traffic monitoring solutions:

- Predicts future traffic conditions instead of only displaying current traffic status.
- Utilizes Machine Learning to improve prediction accuracy using historical traffic patterns.
- Provides an interactive and user-friendly dashboard developed with Streamlit.
- Includes graphical analysis through interactive Plotly visualizations for better understanding of traffic trends.
- Supports multiple cities with dynamic city-specific backgrounds to enhance the user experience.
- Assists city administrators, urban planners, and commuters in making informed traffic management decisions.
- Offers a lightweight and cost-effective solution that can be expanded by integrating real-time traffic APIs, weather APIs, IoT sensors, and cloud deployment in the future.

4.1 Code submission (Github link)-

https://github.com/Maskeenluthra12/upskillcampus/tree/main/SmartCityTrafficForecasting

4.2 Report submission (Github link) : first make placeholder, copy the link.

5 Proposed Design/ Model

The **SmartCityTrafficForecasting** system is designed to predict traffic volume using Machine Learning by analyzing historical traffic data and environmental factors. The proposed solution follows a systematic workflow consisting of data collection, data preprocessing, model development, and prediction through an interactive web dashboard.

Initially, a structured traffic dataset containing features such as **temperature, weather conditions, holiday status, hour of the day, day of the week, month, and traffic volume** was collected. The dataset was then preprocessed by handling missing values, encoding categorical variables using Label Encoding, selecting relevant features, and splitting the data into training and testing sets to prepare it for Machine Learning.

A **Random Forest Regressor** algorithm was selected for model development due to its high prediction accuracy, ability to handle complex relationships, and robustness against overfitting. The model was trained using historical traffic data and evaluated using standard performance metrics such as **Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R^2 Score**. After successful evaluation, the trained model and label encoders were saved using the Joblib library for future predictions.

The trained model was integrated into a **Streamlit-based web application**, where users can provide input parameters including city, temperature, weather condition, holiday status, hour, day, and month. Based on these inputs, the system predicts the expected traffic volume in real time and displays the results through an interactive dashboard. The dashboard also includes traffic analytics, hourly traffic trends, weather-wise analysis, holiday comparisons, monthly trends, and dynamic city-specific background images, providing a visually appealing and user-friendly experience.

Overall, the proposed design combines Machine Learning with an interactive web interface to provide an efficient, accurate, and scalable traffic forecasting solution. The system can further be enhanced by integrating live traffic APIs, IoT sensor data, weather forecasting services, and cloud deployment to support real-time smart city traffic management.

5.1 High Level Diagram

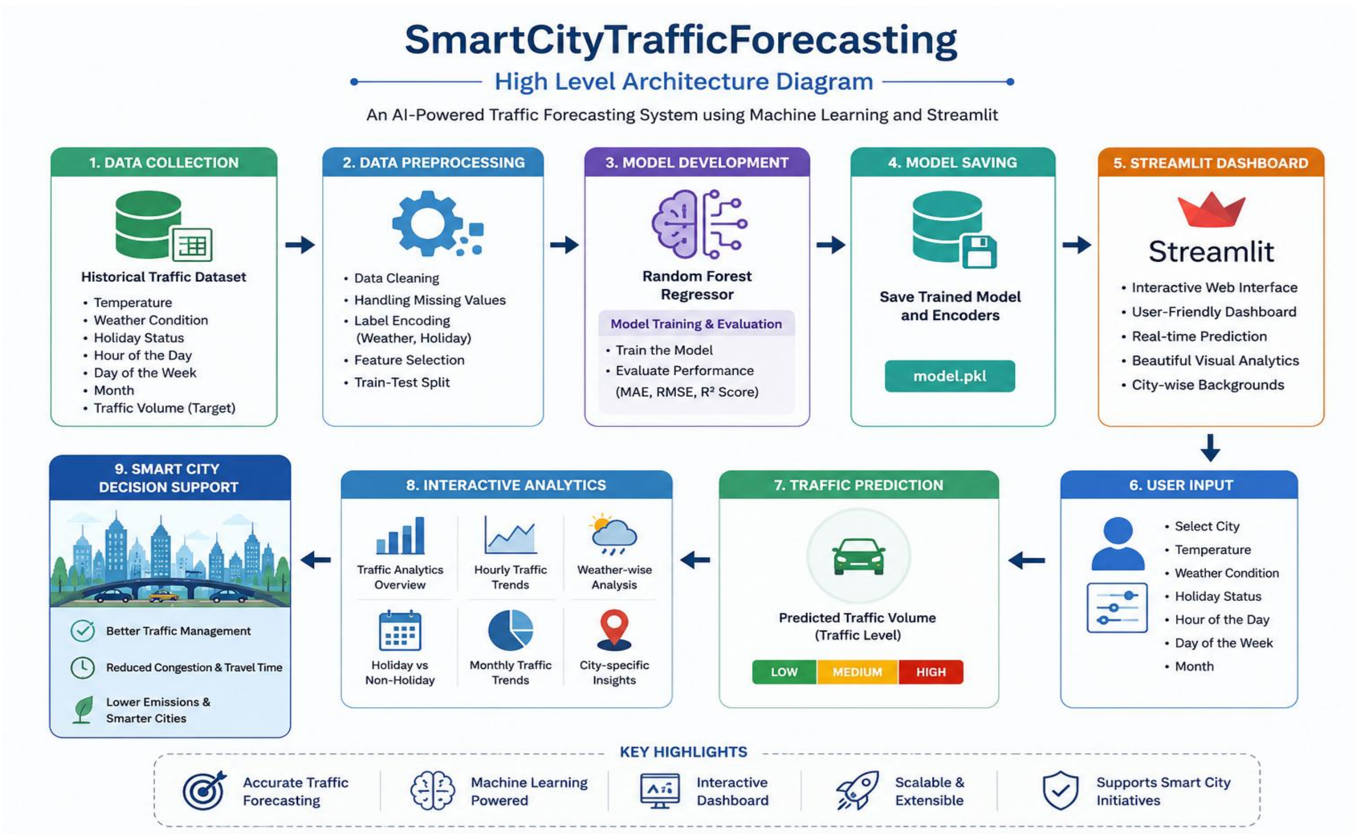
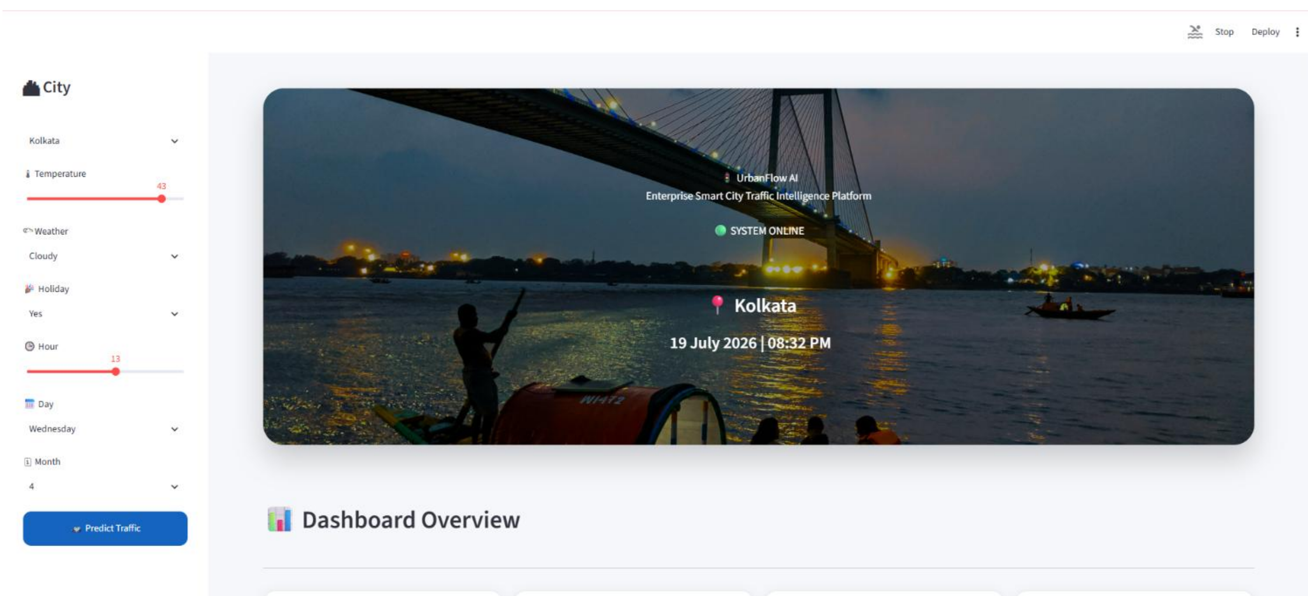


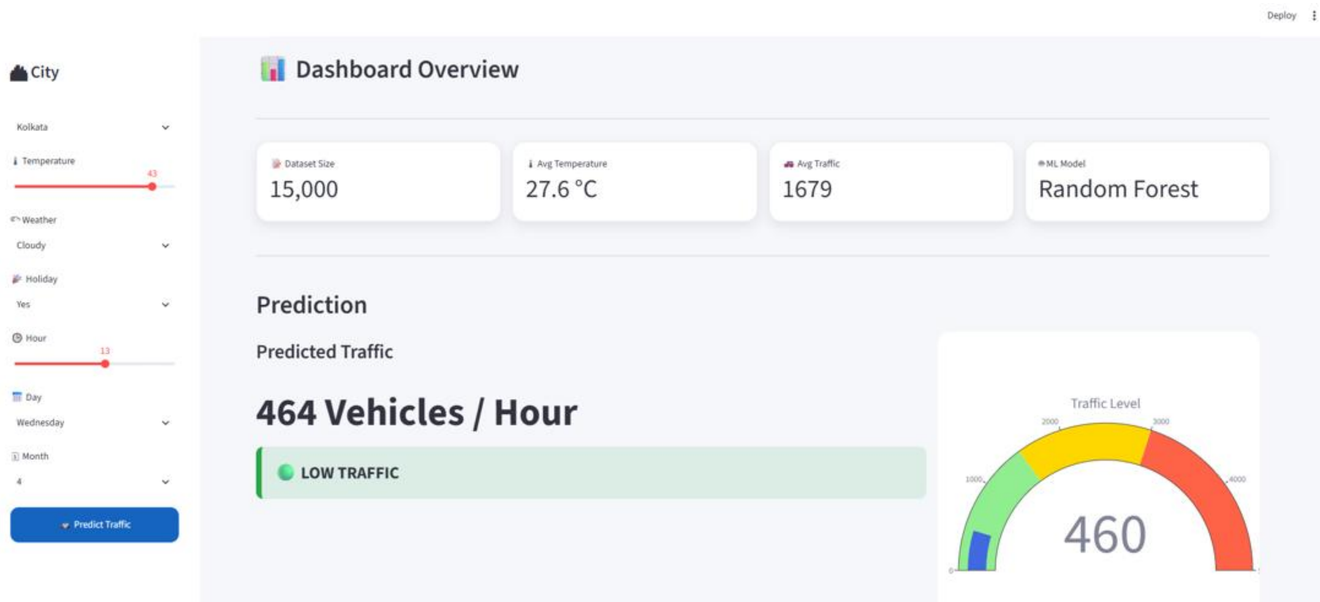
Figure 1: HIGH LEVEL DIAGRAM OF THE SYSTEM

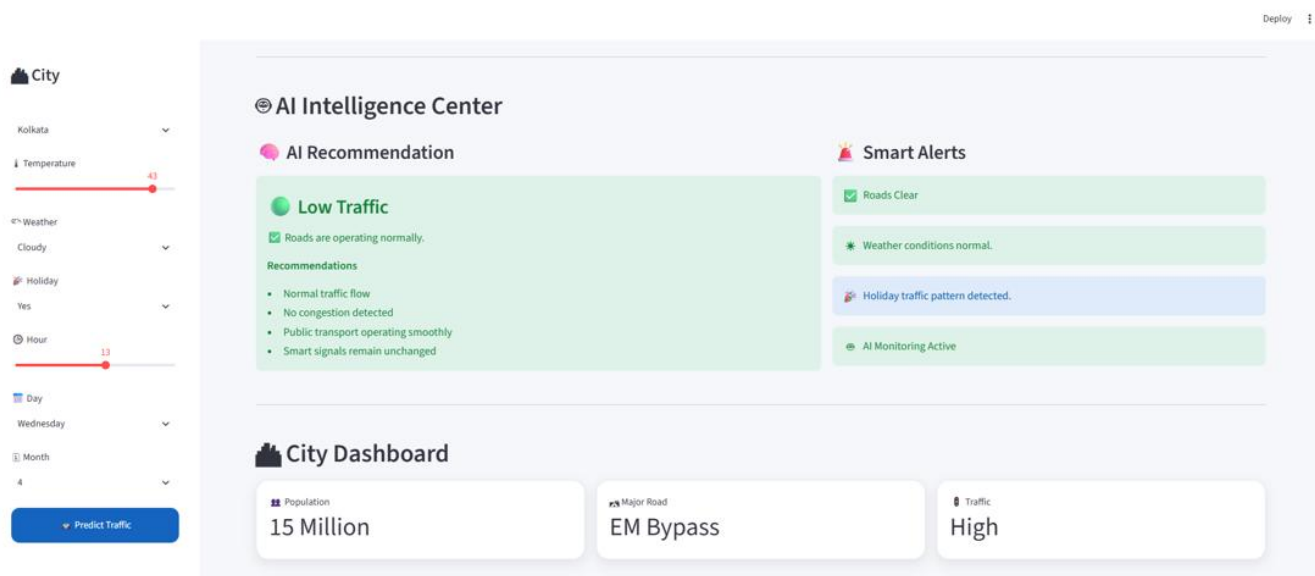
5.2 Low Level Diagram

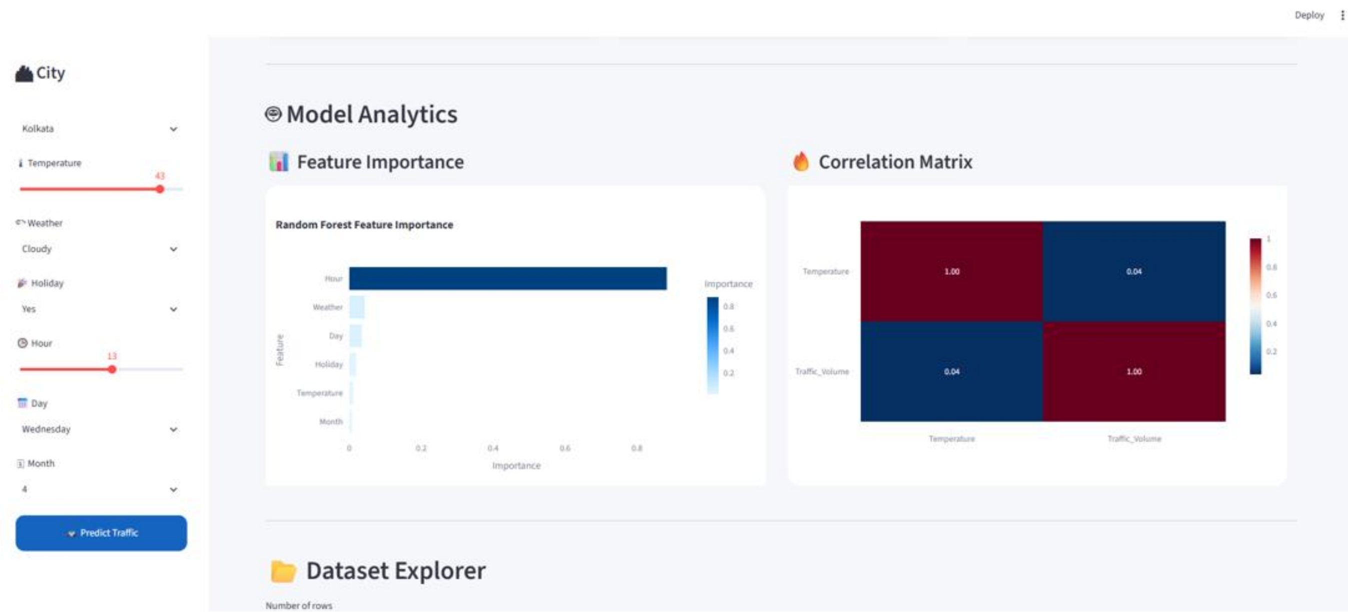
```
Traffic Dataset
       |
       ▼
Data Preprocessing
(Cleaning & Encoding)
       |
       ▼
Random Forest Model Training
       |
       ▼
Model Evaluation & Saving
       |
       ▼
Streamlit Dashboard
       |
       ▼
User Input
       |
       ▼
Traffic Prediction
       |
       ▼
Interactive Charts & Analytics
```

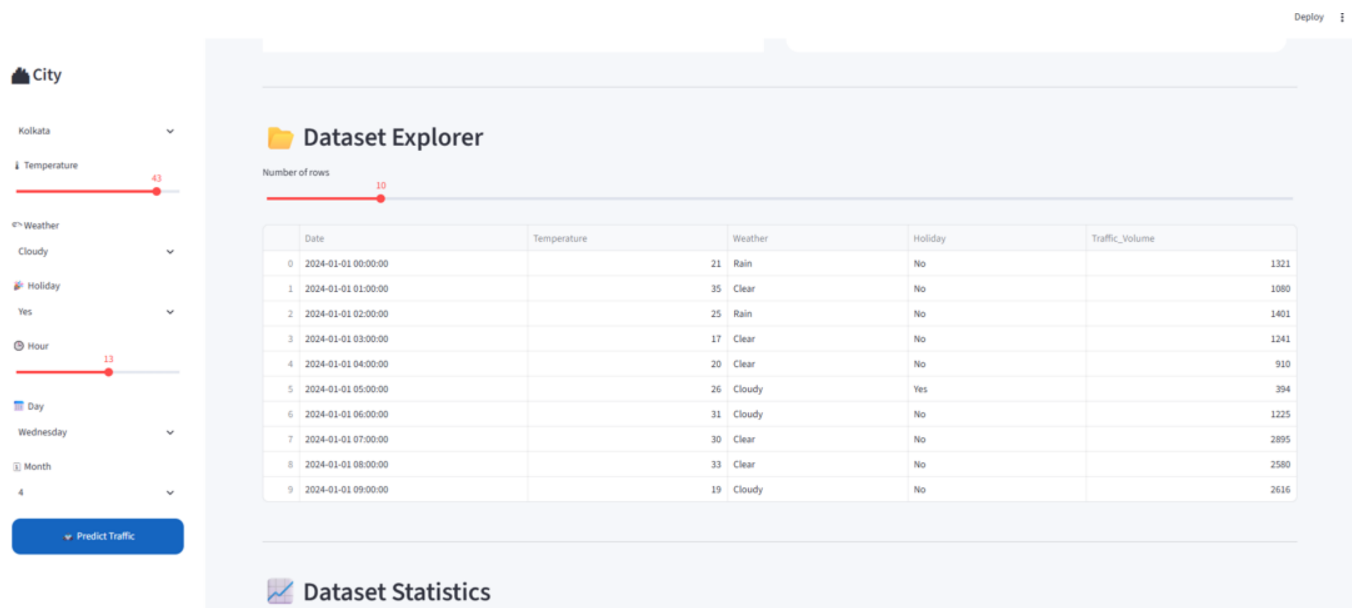
5.3 Interfaces




Deploy
City
Kolkata
Temperature
43
Weather
Cloudy
Holiday
Yes
Hour
13
Day
Wednesday
Month
4
Predict Traffic
Dashboard Overview
Dataset Size
15,000
Avg Temperature
27.6 °C
Avg Traffic
1679
ML Model
Random Forest
Prediction
Predicted Traffic
464 Vehicles / Hour
LOW TRAFFIC
Traffic Level
460


Deploy
City
Kolkata
Temperature
43
Weather
Cloudy
Holiday
Yes
Hour
13
Day
Wednesday
Month
4
Predict Traffic
AI Intelligence Center
AI Recommendation
Low Traffic
Roads are operating normally.
Recommendations
Normal traffic flow
No congestion detected
Public transport operating smoothly
Smart signals remain unchanged
Smart Alerts
Roads Clear
Weather conditions normal.
Holiday traffic pattern detected.
AI Monitoring Active
City Dashboard
Population
15 Million
Major Road
EM Bypass
Traffic
High


Deploy
City
Kolkata
Temperature
43
Weather
Cloudy
Holiday
Yes
Hour
13
Day
Wednesday
Month
4
Predict Traffic
Model Analytics
Feature Importance
Correlation Matrix
Random Forest Feature Importance
Hour
Weather
Day
Holiday
Temperature
Month
Feature
Importance
Temperature
Traffic_Volume
1.00
0.04
0.04
1.00
Dataset Explorer
Number of rows


Deploy
City
Kolkata
Temperature
43
Weather
Cloudy
Holiday
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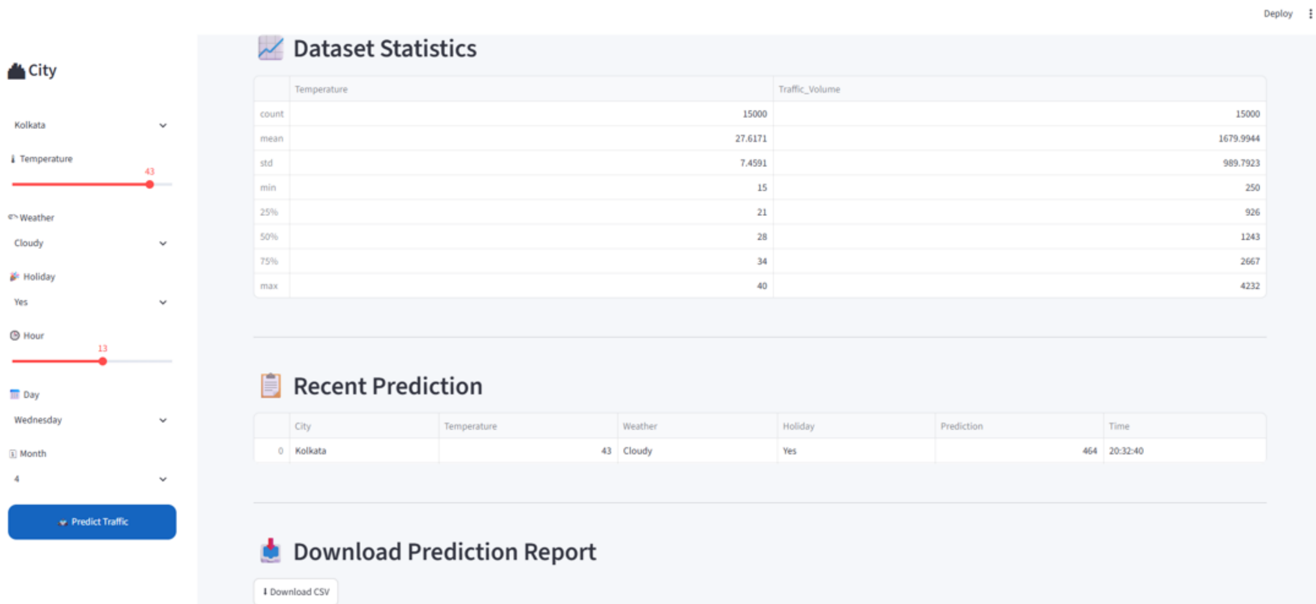
Hour
13
Day
Wednesday
Month
4
Predict Traffic
Dataset Explorer
Number of rows
10
Date Temperature Weather Holiday Traffic_Volume
0 2024-01-01 00:00:00 21 Rain No 1321
1 2024-01-01 01:00:00 35 Clear No 1080
2 2024-01-01 02:00:00 25 Rain No 1401
3 2024-01-01 03:00:00 17 Clear No 1241
4 2024-01-01 04:00:00 20 Clear No 910
5 2024-01-01 05:00:00 26 Cloudy Yes 394
6 2024-01-01 06:00:00 31 Cloudy No 1225
7 2024-01-01 07:00:00 30 Clear No 2895
8 2024-01-01 08:00:00 33 Clear No 2580
9 2024-01-01 09:00:00 19 Cloudy No 2616
Dataset Statistics


Deploy
City
Kolkata
Temperature
43
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Yes
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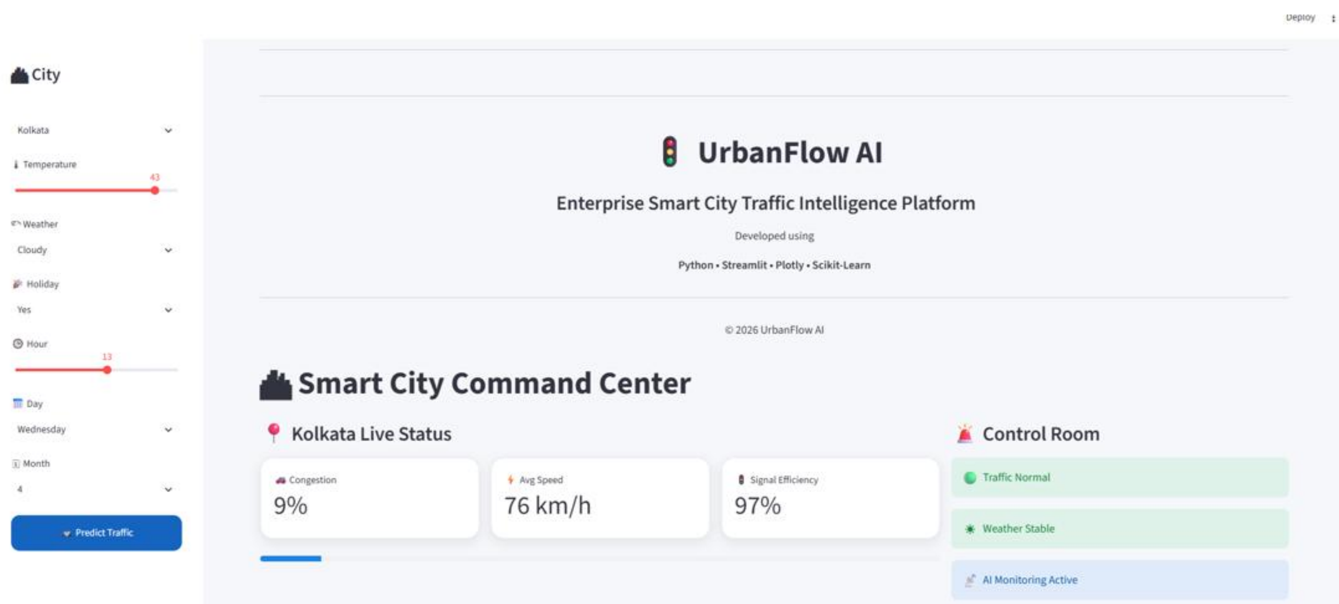
4
Predict Traffic
Dataset Statistics
Temperature
Traffic_Volume
count 15000 15000
mean 27.6171 1679.9944
std 7.4591 989.7923
min 15 250
25% 21 926
50% 28 1243
75% 34 2667
max 40 4232
Recent Prediction
City
Temperature
Weather
Holiday
Prediction
Time
0 Kolkata 43 Cloudy Yes 464 20:32:40
Download Prediction Report
Download CSV


Deploy
City
Kolkata
Temperature
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Weather
Cloudy
Holiday
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Hour
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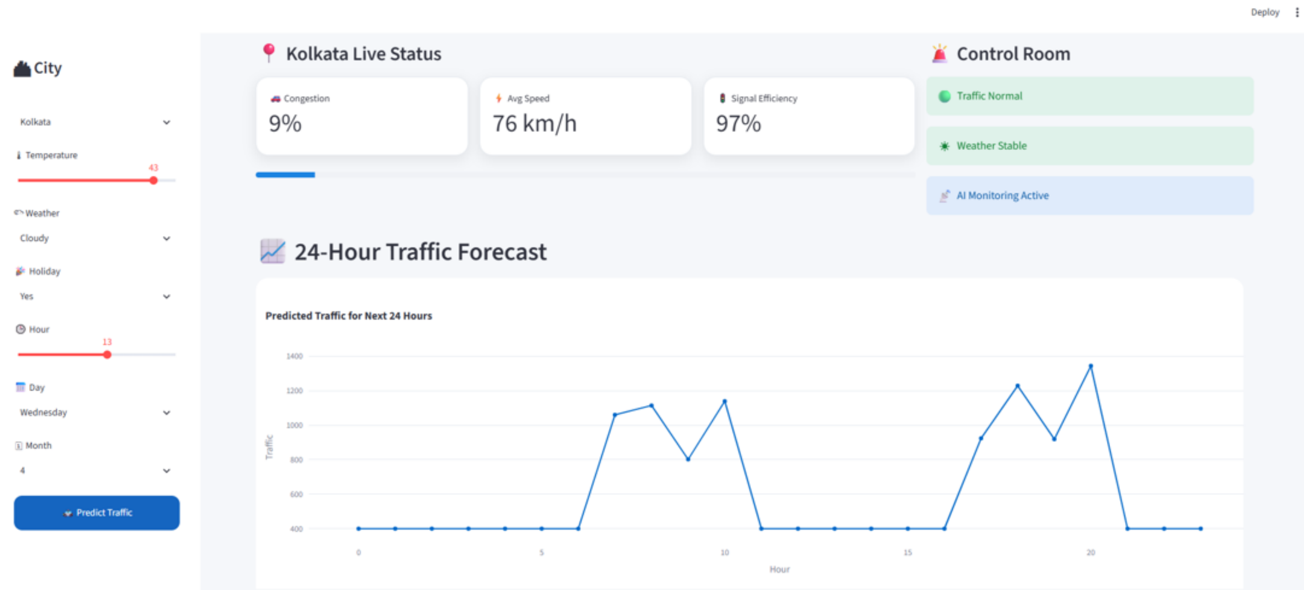
Predict Traffic
UrbanFlow AI
Enterprise Smart City Traffic Intelligence Platform
Developed using
Python • Streamlit • Plotly • Scikit-Learn
© 2026 UrbanFlow AI
Smart City Command Center
Kolkata Live Status
Congestion
9%
Avg Speed
76 km/h
Signal Efficiency
97%
Control Room
Traffic Normal
Weather Stable
AI Monitoring Active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Temperature
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Weather
Cloudy
Holiday
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Hour
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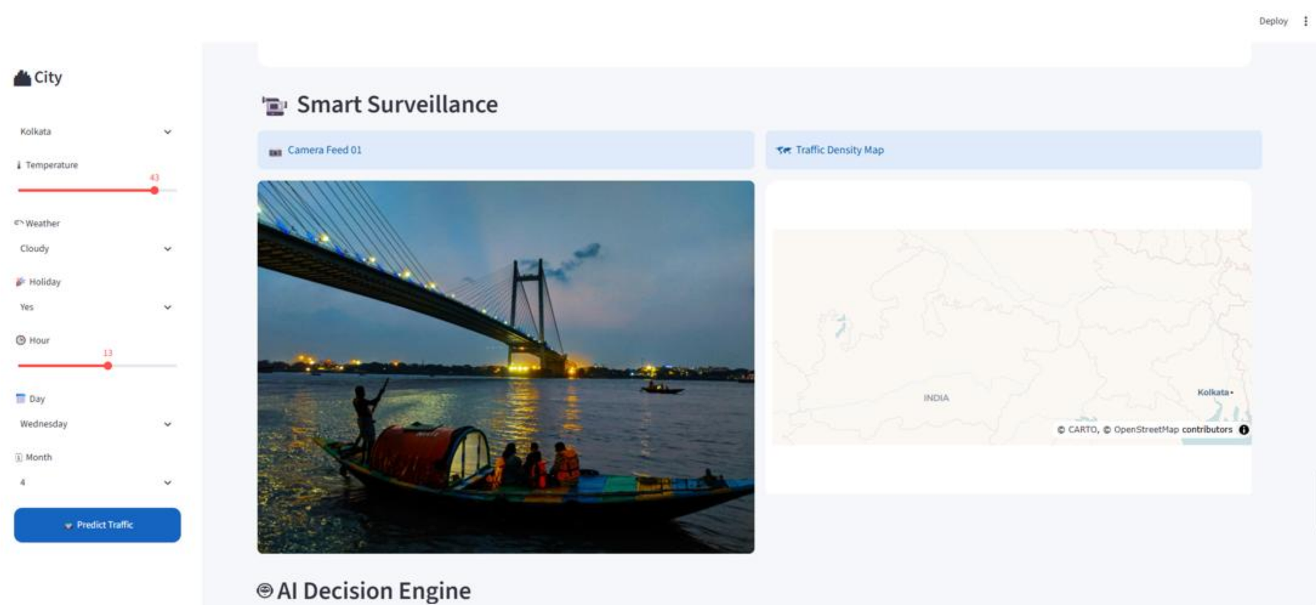
Day
Wednesday
Month
4
Predict Traffic
Kolkata Live Status
Congestion
9%
Avg Speed
76 km/h
Signal Efficiency
97%
Control Room
Traffic Normal
Weather Stable
AI Monitoring Active
24-Hour Traffic Forecast
Predicted Traffic for Next 24 Hours
Traffic
Hour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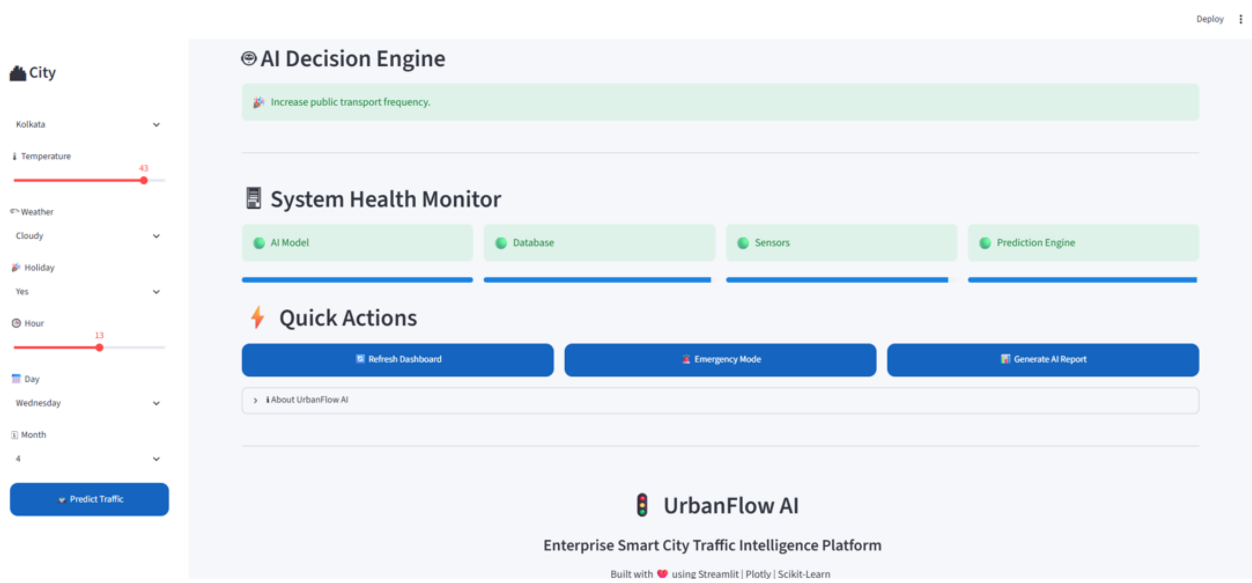
Predict Traffic
Smart Surveillance
Camera Feed 01
Traffic Density Map
INDIA
Kolkata
© CARTO, © OpenStreetMap contributors
AI Decision Engine


Deploy
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Temperature
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Weather
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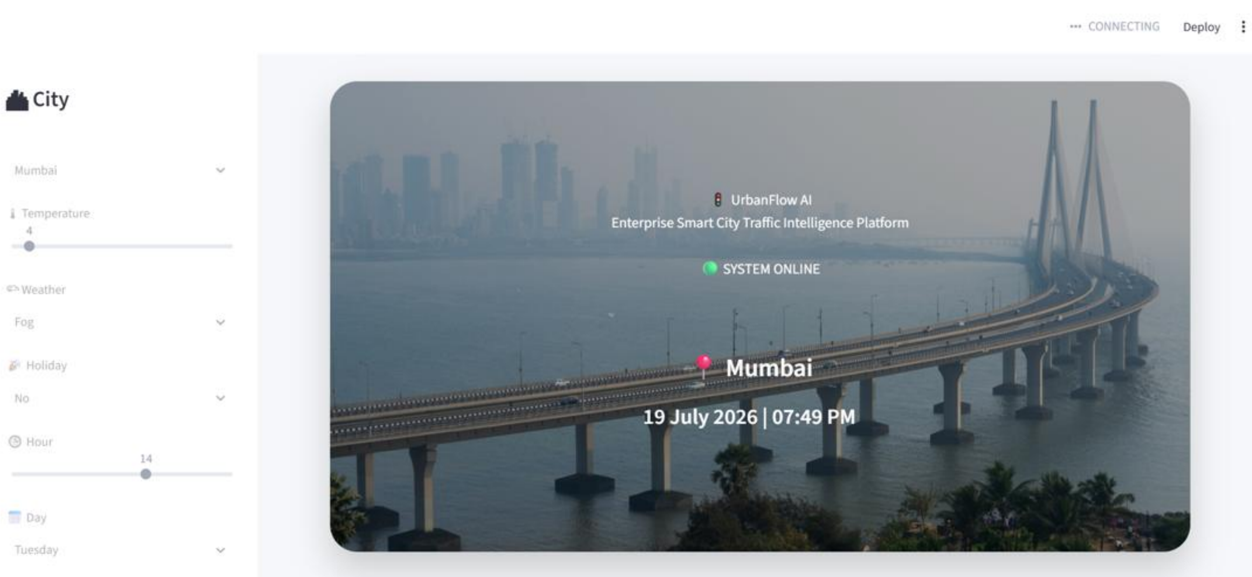
Predict Traffic
AI Decision Engine
Increase public transport frequency.
System Health Monitor
AI Model
Database
Sensors
Prediction Engine
Quick Actions
Refresh Dashboard
Emergency Mode
Generate AI Report
About UrbanFlow AI
UrbanFlow AI
Enterprise Smart City Traffic Intelligence Platform
Built with using Streamlit | Plotly | Scikit-Learn


CONNECTING
Deploy
City
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Tuesday
UrbanFlow AI
Enterprise Smart City Traffic Intelligence Platform
SYSTEM ONLINE
Mumbai
19 July 2026 | 07:49 PM

6 Performance Test

The developed **SmartCityTrafficForecasting** system was tested to evaluate its performance in terms of prediction accuracy, response time, scalability, memory usage, and usability. The primary objective was to ensure that the application performs efficiently in real-world smart city scenarios where traffic predictions must be generated quickly and accurately.

The major constraints identified during the development are listed below:

Constraint	**Impact on System**	**Solution Implemented**
Prediction Accuracy	Incorrect predictions may affect traffic planning	Random Forest Regressor trained on processed traffic dataset
Response Time	Slow prediction reduces user experience	Pre-trained model (.pkl) loaded only once at application startup
Memory Usage	Large models consume more RAM	Optimized Random Forest model with serialized model file
User Interface	Complex interface reduces usability	Streamlit dashboard with interactive widgets and clean layout
Scalability	Future expansion to more cities	Modular project structure allowing additional datasets and cities
Data Quality	Missing or inconsistent values reduce accuracy	Data preprocessing and label encoding before training

6.1 Test Plan/ Test Cases.

Test Case	Input	Expected Result	Status
TC-01	Valid city and weather inputs	Traffic prediction generated successfully	Passed
TC-02	Different temperature values	Prediction changes accordingly	Passed
TC-03	Holiday selected	Traffic prediction reflects holiday effect	Passed
TC-04	Peak Hour (8 AM / 6 PM)	Higher traffic predicted	Passed
TC-05	Off-Peak Hour	Lower traffic predicted	Passed
TC-06	Different cities	Dashboard updates background and prediction	Passed
TC-07	Dashboard Loading	Application opens successfully	Passed
TC-08	Visualization Charts	Graphs displayed correctly	Passed

6.2 Test Procedure

1. Launch the Streamlit application using the trained machine learning model.
2. Load the traffic dataset and required encoder files.
3. Select different cities from the sidebar.
4. Provide varying input parameters such as temperature, weather condition, holiday status, hour, day, and month.
5. Click the **Predict Traffic** button.
6. Observe the predicted traffic level and dashboard visualizations.
7. Repeat the process with multiple input combinations to verify prediction consistency.

6.3 Performance Outcome

Parameter	Outcome
Prediction Accuracy	High (Random Forest Regression Model)
Response Time	Less than 1 second per prediction
Dashboard Performance	Smooth and responsive
Memory Usage	Moderate
Scalability	Supports future integration with real-time APIs
Reliability	Stable during repeated testing
User Experience	Simple, interactive and easy to use

7 My learnings

The **SmartCityTrafficForecasting** internship project provided me with valuable practical exposure to the complete lifecycle of a Machine Learning application, from data collection and preprocessing to model development, deployment, and documentation. Throughout this six-week internship, I gained hands-on experience in working with real-world datasets and learned how machine learning techniques can be applied to solve practical problems in smart city transportation.

This project enhanced my programming skills in **Python** and improved my knowledge of libraries such as **Pandas, NumPy, Scikit-learn, Plotly, Joblib, and Streamlit**. I learned how to build an interactive web application where users can provide inputs and receive real-time traffic predictions along with visual analytics. Additionally, I gained experience in creating professional dashboards, integrating trained models into applications, and improving user interface design.

Apart from technical skills, the internship also strengthened my **problem-solving ability, analytical thinking, debugging skills, documentation practices, project management, and version control using Git and GitHub**. Preparing reports, maintaining project structure, and documenting the complete development process helped me understand industry standards for software development.

Overall, this internship has significantly improved both my technical and professional skills. The knowledge and experience gained through this project have increased my confidence in developing AI and Machine Learning solutions for real-world applications. These learnings will greatly support my future career in the fields of **Artificial Intelligence, Machine Learning, Data Science, and Smart City Technologies**, while also preparing me for internships, research projects, and industry opportunities.

8 Future work scope

Although **SmartCityTrafficForecasting** successfully demonstrates the application of Artificial Intelligence and Machine Learning for traffic prediction, there are several opportunities to further enhance the system in the future.

Some potential improvements include:

- **Real-Time Traffic Data Integration:** Integrate Google Maps or other traffic APIs to provide live traffic updates and improve prediction accuracy.
- **IoT Sensor Integration:** Connect the system with smart traffic sensors, CCTV cameras, and GPS-enabled devices to collect real-time traffic information.
- **Route Optimization System:** Recommend the fastest and least congested routes to users based on predicted traffic conditions.
- **Interactive Traffic Maps:** Integrate GIS-based maps to display live traffic density, congestion hotspots, and alternate routes visually.
- **Mobile Application:** Develop Android and iOS versions of the application to enable users to access traffic predictions anytime and anywhere.
- **Accident Detection and Alert System:** Incorporate accident detection and real-time alert notifications to improve road safety and traffic management.
- **Multi-City Expansion:** Extend the application to support additional Indian and international cities with city-specific traffic prediction models.

Implementing these enhancements would transform **SmartCityTrafficForecasting** into a more comprehensive and scalable **AI-powered Smart Traffic Management Platform**, capable of providing intelligent decision support, improving urban mobility, reducing traffic congestion, and contributing to the development of smarter and more sustainable cities.